

DEPARTMENT OF MATHEMATICS
NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

Program: B. Tech. Semester-IV
Course title: Applied Linear Algebra
Course Code: MAIC-204

Time : 02:00 PM-05:00 PM
M.M.: 50

NOTE: Attempt all the Questions. Each question carries equal marks.

1. Let V be a vector space spanned by a finite set of vectors $\beta_1, \beta_2, \dots, \beta_m$. Prove that any independent set of vectors in V is finite and contains no more than m elements.
2. In $\mathbb{R}^3$, let $\alpha_1 = (1, 0, 1)$, $\alpha_2 = (0, 1, -2)$, $\alpha_3 = (-1, -1, 0)$. Check if f is a linear functional on $\mathbb{R}^3$ such that $f(\alpha_1) = 1$, $f(\alpha_2) = -1$, $f(\alpha_3) = 3$ and if $\alpha = (a, b, c)$ find $f(\alpha)$.
3. State and prove the Rank Nullity Theorem.
4. Consider the vectors
 $\beta_1 = (3, 0, 4)$, $\beta_2 = (-1, 0, 7)$, $\beta_3 = (2, 9, 11)$
in $\mathbb{R}^3$ equipped with the standard inner product space. Find out the orthonormal set of vectors $\{\alpha_1, \alpha_2, \alpha_3\}$ from these vectors using Gram Schmidt process.
5. Let T be a linear operator on $\mathbb{R}^3$ defined by
 $(x, y, z) \mapsto (3x, -y, 2x + y + z)$.
Is T invertible? If so, find a rule for T^{-1} like the one which defines T .
6. Let V be $\mathbb{C}^2$, with the standard inner product space. Let T be the linear operator on V which is represented in the standard ordered basis by the matrix:

$$\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}.$$

Show that T is normal and find an orthonormal basis for V , consisting of characteristic vectors for T .

7. Solve the following set of linear equations by LU-decomposition method:

$$\begin{aligned} 1x + 2y + 4z &= 7, \\ 3x + 8y + 14z &= 23, \\ 2x + 6y + 13z &= 22. \end{aligned}$$

8. Define norm in an inner product space. Also, derive the polarization identities.
9. Find the nature of the quadratic form:
 $x^2 + 4y^2 + z^2 + 1xy - 1yz + 6zx$.
10. If $C(x) = (x - 3)^3(x - 2)^4$ and $M(x) = (x - 3)^2(x - 2)^2$. Then find all the possible Jordan Canonical form of the given characteristic and minimal polynomials.

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